get_family_trait_summary.R

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`get_family_trait_summary` *Get Family-Level Trait Summary for a Selected Taxon*

Description

Looks up the taxonomic family of ‘sel_taxon‘ in ‘data_classification‘, collects all same-family taxa from ‘data_traits_raw‘ for the chosen trait domain, and returns a summary tibble with per-taxon statistics (‘n‘, ‘min‘, ‘q25‘, ‘median‘, ‘mean‘, ‘q75‘, ‘max‘) sorted by ‘median‘.

Usage

```
get_family_trait_summary(
  data_traits_raw,
  data_classification,
  sel_taxon,
  sel_domain,
  sel_rank = "family",
  verbose = TRUE
)
```

Arguments

`data_traits_raw`
: A data frame of raw trait observations. Must contain columns ‘taxon_name‘ (character), ‘trait_domain_name‘ (character), and ‘trait_value‘ (numeric).

`data_classification`
: A data frame mapping taxon names to taxonomic ranks. Must contain columns ‘sel_name‘ (character) and ‘family‘ (character).

`sel_taxon`
: Character scalar. Name of the focal taxon to look up.

`sel_domain`
: Character scalar. Trait domain to summarise (matched against ‘trait_domain_name‘ in ‘data_traits_raw‘).

sel_rank — Character scalar. Name of the taxonomic rank column in ‘data_classification‘ to use for grouping (e.g. ‘"family"‘, ‘"genus"‘, ‘"order"‘). Defaults to ‘"family"‘.

verbose — Logical. If ‘TRUE‘ (default), progress messages are printed to the console via ‘cli‘.

Details

1. The function joins ‘sel_taxon‘ against ‘data_classification‘ to resolve the value in the ‘sel_rank‘ column. 2. All taxa sharing that same ‘sel_rank‘ value are identified from ‘data_classification‘. 3. ‘data_traits_raw‘ is filtered to those taxa and ‘sel_domain‘, then grouped by ‘taxon_name‘ to produce summary statistics. 4. If the rank value cannot be resolved (taxon absent from ‘data_classification‘ or the ‘sel_rank‘ column is ‘NA‘), only ‘sel_taxon‘’s own observations are summarised.

Value

A tibble with one row per taxon that has data for ‘sel_domain‘ within the same ‘sel_rank‘ group as ‘sel_taxon‘. Columns: ‘taxon_name‘, ‘n‘ (integer observation count), ‘min‘, ‘q25‘, ‘median‘, ‘mean‘, ‘q75‘, ‘max‘ (all numeric). Rows are sorted ascending by ‘median‘. If ‘sel_taxon‘ is not found in ‘data_classification‘, only observations for ‘sel_taxon‘ itself are summarised.

See Also

[generate_trait_qc_report()], [plot_trait_group_distribution()]

Examples

```
## Not run:
data_aux_classification <-
  readr::read_csv(
    here::here("Data/Input/aux_classification_table.csv"),
    show_col_types = FALSE
  )

data_family_comparison <-
  get_family_trait_summary(
    data_traits_raw = data_traits_raw,
    data_classification = data_aux_classification,
    sel_taxon = "Anacyclus clavatus",
    sel_domain = "Leaf Area"
  )

## End(Not run)
```

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